

M L P R O J E C T S

# Airline & Aviation Analytics Portfolio

Five end-to-end machine learning projects spanning route profitability, delay prediction, CO<sub>2</sub> forecasting, customer churn and satisfaction modelling.

CO<sub>2</sub> Emissions

Route Profitability

Delay DNN

Churn

Satisfaction



# CO<sub>2</sub> Emissions from Aviation

Enhanced ML & Forecasting Pipeline | ESPRIT 3ALINFO



## Objective

- Forecast annual CO<sub>2</sub> emissions by country from aviation data
- Compare regression, time-series and ensemble models
- 10-year forward projection + anomaly detection



## Data Cleaning

- Rename & standardize columns (Entity→Country, CO<sub>2</sub> field)
- Drop nulls on required fields; deduplicate by Country+Year
- Parse Year as int; sort by Country & Year



## Models Implemented

- **Linear / Ridge / Lasso / ElasticNet**
- **Random Forest (tuned)**
- **Gradient Boosting / XGBoost / LightGBM**
- **SVR & KNN**
- **ARIMA + Holt-Winters + Prophet**

Baseline regression & regularization

Non-linear ensemble with hyperparameter search

Boosted trees for high accuracy

Support vector and proximity-based regression

Univariate time-series forecasting



## Feature Engineering

- Lag features (1, 2, 3 years)
- Rolling avg & std (3Y, 5Y)
- YoY growth rate & momentum
- Log-transform for skewness
- Year index + quadratic trend

9+

Models Compared

10 yrs

Horizon Forecast

R<sup>2</sup> / MAE

Key Metric

ARIMA+Prophet

Techniques

# Airline Route Profitability & Cost Analysis

Comprehensive Financial Study — Dubai (DXB) Hub | ~7,500 Flights • 30 Routes

## Objective

- Predict profit margin (regression) per flight
- Classify routes as profitable vs unprofitable
- Identify cost drivers with SHAP explainability
- Route clustering & what-if business simulator

## Data Cleaning & Engineering

- Verified logical consistency:  $\text{Load\_Factor} = \text{Passengers} / \text{Capacity}$
- Cross-checked stored vs computed Profit & Profit\_Margin
- Parsed Flight\_Date → Month, Quarter, DayOfWeek
- Engineered CASK / RASK (cost & revenue per available seat-km)
- Derived Revenue & Cost per Passenger; Is\_Profitable flag

## Models & Techniques

### Regression (Profit)

- Ridge Regression (baseline)
- Random Forest Regressor
- Gradient Boosting Regressor
- LightGBM (best performer)

### Classification (Profitable?)

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting
- Cross-validation (KFold)

### Unsupervised / Explainability

- KMeans Route Clustering
- PCA dimensionality reduction
- SHAP feature importance

## EDA Highlights

- Fleet & route network overview (aircraft type, destinations)
- Revenue / cost / profit distributions

~7,500

Flights

30

Routes

14

Cost Categories

CASK/RASK·R<sup>2</sup>

Key Metric

# Airline Delay Cause — Deep Neural Network

Binary Classification · Validation Accuracy 99%

## Objective

- Predict whether a flight experiences severe weather delay (>100 min)
- Binary classification: WDCase = 1 (severe) / 0 (normal)
- Achieve maximum accuracy using a custom deep neural network

## Data Cleaning

- Dropped non-numeric ID columns (carrier, airport names)
- Removed rows with missing values (dropna)
- Created binary target: weather\_delay > 100 → WDCase = 1
- 75/25 train-test split with shuffle & fixed random state

## DNN Architecture

|                     |                                    |
|---------------------|------------------------------------|
| Input (17 features) | All numerical delay cause features |
| Dense 8 — Tanh      | Initial feature compression        |
| Dense 128 — Sigmoid | Non-linear representation          |
| Dense 64 — Tanh     | Deeper abstraction                 |
| Dense 32 — Tanh     | Feature refinement                 |

## Training Config

- Optimizer: AdamW (lr=0.001, wd=0.004)
- Loss: Binary Cross-Entropy
- Epochs: 100 (Early Stopping, patience=10)
- Batch size: 10,000
- Monitor: val\_accuracy (restore best weights)

**99%**

Validation Accuracy

**5L+Dropout**

Architecture

**AdamW**

Optimizer

**F1 / ROC-AUC**

Metrics



# Airline Customer Churn Prediction

Time-Series Binary Classification • Churn = 0 flights for 6 consecutive months



## Objective

- Identify customers likely to stop flying (no flight for 6+ months)
- Time-aware pipeline: monthly customer activity panel
- Business insight: who to target with retention campaigns



## Data Preparation

- Dynamic column detection (loyalty\_number, year, month, flights)
- Built complete monthly panel per customer (fill missing months → 0)
- Forward/back fill for non-numeric categorical fields
- Churn target: next 6 months all zero-flight → churn=1
- Dropped rows without a full future prediction window



## Feature Engineering

- Recency: months since last positive flight
- Frequency: rolling avg flights (3m, 6m windows)
- Trend: rolling slope for flights & points (6m)
- Engagement: cumulative points & redemption ratio
- Activity drops: recent vs previous windows
- Customer-level aggregates (lifetime metrics)
- Dynamic lag & rolling features for numeric columns



## Models

- **Logistic Regression** Fast linear baseline
- **Random Forest** Ensemble; feature importance
- **XGBoost** Best performer typically

**6-month gap**

Churn Definition

**Time Series**

Data Format

**3 Classifiers**

Models

**RFM signals**

Key Features

# ★ Airline Customer Satisfaction Modeling

Multi-class Classification + Clustering + SHAP Explainability

## 🎯 Objective

- Predict passenger satisfaction from survey & flight data
- Compare LR, Random Forest, and XGBoost/LightGBM
- Explain predictions with SHAP feature importances
- Segment customers with KMeans; build churn proxy label

## 🔧 Preprocessing

- Dynamic schema detection (target column auto-detected)
- Removed ID-like columns (uniqueness ratio > 98%)
- Median imputation for numerics; mode for categoricals
- OneHotEncoding for categoricals; StandardScaler for numerics
- Engineered: total\_delay, delay\_flag, age×distance interaction
- Model-based feature selection using Random Forest importance

## 🤖 Models & Analysis Pipeline

### Classification Models

- Logistic Regression (baseline)
- Random Forest Classifier
- XGBoost / LightGBM (best)
- Stratified 80/20 split

### Explainability

- SHAP values per prediction
- Feature importance ranking
- Delay impact analysis (dep + arr)
- Class-level SHAP beeswarm

### Segmentation & Extras

- KMeans customer clustering
- TruncatedSVD for dim. reduction
- Churn proxy label creation
- Multiclass ROC-AUC scoring

**Satisfaction**

Target

**LR+RF+XGBoost**

Models

**SHAP**

Explainability

**Clustering+Churn**

Extra

# Portfolio Summary — Five ML Projects at a Glance

## CO<sub>2</sub> Emissions Forecasting

Goal: Forecast aviation CO<sub>2</sub> per country

### Data

Annual emissions by country · multi-country

### Cleaning

Column rename, dedup, null drop, type cast

### Models

LR, Ridge, RF, GBM, XGBoost, ARIMA, Prophet, Ensemble



## Route Profitability Analysis

Goal: Predict profit; classify profitable routes

### Data

~7,500 flights · 30 routes · 14 cost columns

### Cleaning

Logic checks (Load Factor, Profit), CASK/RASK engineering

### Models

Ridge, RF, GBM, LightGBM (reg) + classifiers + KMeans



## Delay Cause DNN

Goal: Classify severe weather delay (>100 min)

### Data

Airline delay cause dataset · binary target

### Cleaning

Drop ID cols, remove nulls, create binary WDCase

### Models

Deep Neural Net (5 Dense + Dropout) · AdamW optimizer



## Customer Churn Prediction

Goal: Predict churners: 6 months zero flights

### Data

Monthly loyalty activity · time-series panel

### Cleaning

Complete monthly panel; 0-fill missing months; dynamic cols

### Models

Logistic Regression, Random Forest, XGBoost



## Customer Satisfaction Modeling

Goal: Predict satisfaction; cluster passengers

### Data

Survey + flight data · dynamic schema detection

### Cleaning

Auto schema detection, ID-col removal, delay engineering

### Models

LR, RF, XGBoost/LightGBM + KMeans + SHAP



All projects share a common pattern: Data Cleaning → Feature Engineering → Model Training → Evaluation → Interpretation



# Key Insights & Common Patterns



## Shared ML Methodology

- Data Quality First — validation, dedup, null handling before any model
- Feature Engineering — domain-aware (CASK, lag features, RFM metrics)
- Multiple Models Compared — from linear baselines to boosted ensembles
- Robust Evaluation — accuracy, F1, ROC-AUC, confusion matrix,  $R^2$
- Explainability — SHAP values used across profitability & satisfaction projects
- Business Impact — each project produces actionable output (simulator, clustering, churn targeting)



## Standout Results

- ◆ **99% Accuracy**  
DNN delay classifier achieves near-perfect classification
- ◆ **10-Year CO<sub>2</sub> Forecast**  
Ensemble + Prophet for long-range aviation emission planning
- ◆ **SHAP Explainability**  
Route profitability & satisfaction decisions are interpretable
- ◆ **RFM Churn Model**  
Time-series panel with recency, frequency & trend signals
- ◆ **Full EDA Pipelines**  
Every project includes rich visual exploratory analysis



## Technology Stack

| Data           | Visualization        | ML (Sklearn)                       | Boosting           | Deep Learning              | Time Series                    | Explainability |
|----------------|----------------------|------------------------------------|--------------------|----------------------------|--------------------------------|----------------|
| Pandas · NumPy | Matplotlib · Seaborn | RF · LR · GBM · SVR · KMeans · PCA | XGBoost · LightGBM | TensorFlow / Keras · AdamW | ARIMA · Holt-Winters · Prophet | SHAP           |

*"From raw CSV to business-ready insights — every project follows a rigorous, reproducible ML pipeline."*